

Software Requirements Specification (SRS)

Collaborative Study Material & Quiz Platform

Project: Collaborative Study Material & Quiz Platform

Version: 1.0

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1.0	06-09-2026	Student Team	Initial SRS draft with UML use case diagram embedded	

Approvals

Role	Name	Signature / Email	Date
Course Coordinator			
Project Guide			

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1. Introduction

1.1 Purpose

This document is a Software Requirements Specification (SRS) for the Collaborative Study Material & Quiz Platform. It defines the functional and non-functional requirements, interfaces, and verification criteria for a web-based platform that allows students and instructors to share study material, collaboratively build and take quizzes, discuss topics, and track learning progress.

1.2 Scope

The platform covers user registration and authentication, upload and organisation of study material (notes, slides, PDFs, links), creation and administration of quizzes, auto-graded quiz attempts, peer discussion and rating of material, leaderboards and progress analytics, and administrative moderation of content and users. It excludes offline/desktop clients, payment processing, and integration with external Learning Management Systems (LMS) beyond optional single sign-on.

1.3 Audience

Developers, QA Engineers, UI/UX Designers, Students, Instructors, Platform Administrators, and Assessment Evaluators.

1.4 Definitions

List of acronyms: SRS, UI, API, JWT, OTP, TLS, RBAC, RTO, RPO, LMS, SLA, WCAG.

2. Overall description

2.1 Product perspective

The Collaborative Study Material & Quiz Platform is a standalone, cloud-hosted web application accessible via desktop and mobile browsers. It integrates with an external Authentication Service for identity verification, a Notification Service for reminders and alerts, and a Plagiarism Detection Service for scanning uploaded material and quiz content.

2.2 Major product functions (detailed)

- Register / log in and manage user profile
- Upload, tag, version, and search study material
- Create, edit, and publish quizzes from a shared question bank
- Attempt quizzes with auto-grading and instant feedback
- Comment on, discuss, and rate shared study material
- View leaderboards, scores, and personal progress analytics
- Receive notifications for new material, quiz deadlines, and results
- Administer users, moderate flagged content, and review plagiarism reports

2.3 User roles and characteristics (expanded)

- Student: uses the platform to browse material, take quizzes, discuss topics, and track progress; expects a fast, intuitive UI.
- Instructor: uploads material, authors quizzes, monitors class performance, and moderates discussions for their groups.
- Administrator: manages accounts, roles, and platform-wide moderation; requires audit trails and bulk-action tools.

2.4 Operating environment

Modern web browsers (Chrome, Firefox, Safari, Edge) on desktop and mobile; cloud-hosted backend (containerised microservices) with a managed relational database and object storage for uploaded files; horizontally scalable behind a load balancer.

2.5 Constraints

Compliance with data-protection regulations (e.g. GDPR-style consent and data-deletion), accessibility conformance (WCAG 2.1 AA), use of approved cryptographic libraries, and dependency on third-party plagiarism-detection and notification APIs.

3. External interface requirements

3.1 User interfaces

Primary UI: responsive web interface with a dashboard, material library, quiz builder/player, discussion threads, and leaderboard views. Accessibility: screen-reader friendly labels, keyboard navigation, and a high-contrast mode.

3.2 Hardware interfaces

- Standard client devices (desktop, laptop, tablet, smartphone) with camera/microphone optional for proctored quizzes
- No specialised hardware required on the server side beyond standard cloud compute and storage

3.3 Software interfaces

- Authentication Service API (OAuth2 / JWT over TLS) for sign-up, login, and session management
- Notification Service API (email / push) for reminders and result alerts
- Plagiarism Detection Service API for scanning uploaded material and quiz submissions
- Object storage API for storing and retrieving uploaded files

3.4 Communications

- TLS 1.2+ enforced for all client-server and service-to-service communication
- Retry/backoff strategy for third-party service calls; queued processing for large file uploads and plagiarism scans

4. System features (detailed)

Each requirement below includes acceptance criteria and a reference test case. IDs follow CSQ-F-####.

4.1 User Registration & Authentication

Description: Allow students and instructors to register, verify their identity, and securely log in and out of the platform.

Req ID	Requirement	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
CSQ-F-001	The system shall allow a new user to register with an email address, password, and role selection (Student/Instructor).	Functional	High	Student / Instructor	AC-CSQ-F-001: Valid details create an active account and send a verification email. Test: TC-Auth-01	Requires Notification Service

Req ID	Requirement	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
CSQ-F-002	The system shall authenticate users via the Authentication Service and issue a signed session token (JWT) on successful login.	Functional	High	Security	AC-CSQ-F-002: Correct credentials return a valid token; incorrect credentials are rejected. Test: TC-Auth-02	Requires Authentication Service
CSQ-F-003	The system shall lock an account for 15 minutes after 5 consecutive failed login attempts and record an audit event.	Functional	High	Security	AC-CSQ-F-003: After 5 failures, further attempts are blocked and a message is displayed. Test: TC-Auth-03	Audit log required

4.2 Study Material Management

Description: Enable instructors and students to upload, organise, search, and version study material.

Req ID	Requirement	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
CSQ-F-010	The system shall allow an instructor to upload study material (PDF, slides, links) tagged with subject, topic, and difficulty.	Functional	High	Instructor	AC-CSQ-F-010: Uploaded file is stored, tagged, and visible in the library within 30s. Test: TC-Mat-01	Requires object storage
CSQ-F-011	The system shall run every uploaded document through the Plagiarism Detection Service before publishing.	Functional	High	Academic Integrity	AC-CSQ-F-011: Material above the similarity threshold is flagged for review before publication. Test: TC-Mat-02	Depends on Plagiarism Service
CSQ-F-012	The system shall allow students to search and filter material by subject, topic, rating, and upload date.	Functional	Medium	Student	AC-CSQ-F-012: Search returns matching results within 2s for a library of 10,000 items. Test: TC-Mat-03	Search index required
CSQ-F-013	The system shall retain previous versions of a material item whenever it is updated.	Functional	Medium	Instructor	AC-CSQ-F-013: Version history is viewable and previous versions are downloadable. Test: TC-Mat-04	Versioned storage

4.3 Quiz Creation & Management

Requirements: Instructors shall create quizzes from a shared question bank, set time limits and difficulty, and publish quizzes to specific student groups. IDs: CSQ-F-020..CSQ-F-023.

Req ID	Requirement	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
CSQ-F-020	The system shall allow an instructor to create a quiz from a shared question bank, adding	Functional	High	Instructor	AC-CSQ-F-020: Quiz is saved as a draft with all added	Requires question bank module

	multiple-choice, true/false, and short-answer questions.				questions retrievable. Test: TC-QuizC-01	
CSQ-F-021	The system shall allow an instructor to configure quiz settings including time limit, number of attempts, shuffle order, and difficulty level.	Functional	High	Instructor	AC-CSQ-F-021: Configured settings are enforced during a quiz attempt. Test: TC-QuizC-02	
CSQ-F-022	The system shall allow an instructor to publish a quiz to one or more specific student groups with a defined start and end availability window.	Functional	High	Instructor	AC-CSQ-F-022: Quiz is visible only to selected groups within the configured window. Test: TC-QuizC-03	
CSQ-F-023	The system shall allow an instructor to reuse and edit questions from the shared question bank when creating a new quiz.	Functional	Medium	Instructor	AC-CSQ-F-023: Reused question retains its bank history; edits apply only to the new quiz's copy. Test: TC-QuizC-04	Requires versioned question bank

4.4 Quiz Taking & Auto-Grading

Requirements: Students shall attempt published quizzes within the allotted time; the system shall auto-grade objective questions and display results and correct answers immediately after submission. IDs: CSQ-F-030..CSQ-F-032.

Req ID	Requirement	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
CSQ-F-030	The system shall allow a student to attempt a published quiz within the allotted time limit and configured attempt count.	Functional	High	Student	AC-CSQ-F-030: Attempt is timed and blocked once the time limit or max attempts is reached. Test: TC-Quiz-01	
CSQ-F-031	The system shall auto-grade objective questions (MCQ, true/false) immediately upon quiz submission.	Functional	High	Student	AC-CSQ-F-031: Score is calculated and displayed within 2 seconds of submission. Test: TC-Quiz-02	
CSQ-F-032	The system shall display correct answers and explanations after grading, unless the instructor has disabled answer reveal for that quiz.	Functional	Medium	Instructor / Student	AC-CSQ-F-032: Correct answers are shown only when the reveal setting is enabled. Test: TC-Quiz-03	Depends on CSQ-F-021

4.5 Discussion & Collaboration

Requirements: Users shall comment on study material and quiz questions, reply in threaded discussions, and rate/review material on a 5-point scale. IDs: CSQ-F-040..CSQ-F-042.

Req ID	Requirement	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
CSQ-F-040	The system shall allow a user to post a comment on a study material item or a quiz question.	Functional	Medium	Student / Instructor	AC-CSQ-F-040: Comment appears under the item within 5 seconds and is visible to all users with access. Test: TC-Disc-01	
CSQ-F-041	The system shall support threaded replies to comments, up to 3 levels of nesting.	Functional	Medium	Student / Instructor	AC-CSQ-F-041: Replies render nested and in chronological order. Test: TC-Disc-02	
CSQ-F-042	The system shall allow a student to rate a study material item on a 5-point scale and view the current average rating.	Functional	Low	Student	AC-CSQ-F-042: Average rating recalculates immediately after a new rating is submitted. Test: TC-Disc-03	

4.6 Leaderboard, Analytics & Notifications

Requirements: The system shall maintain per-quiz and overall leaderboards, provide instructors with class performance analytics, and send notifications for new material, upcoming quiz deadlines, and result publication via the Notification Service. IDs: CSQ-F-050..CSQ-F-052.

Req ID	Requirement	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
CSQ-F-050	The system shall maintain a per-quiz leaderboard ranking students by score and completion time.	Functional	Medium	Student	AC-CSQ-F-050: Leaderboard updates within 5 seconds of quiz submission; ties are broken by completion time. Test: TC-Lead-01	
CSQ-F-051	The system shall provide instructors with a class performance dashboard showing average scores, question-level difficulty, and participation rate.	Functional	Medium	Instructor	AC-CSQ-F-051: Dashboard reflects the latest submitted attempts and is exportable as CSV. Test: TC-Lead-02	

CSQ-F-052	The system shall send a notification via the Notification Service for new material, upcoming quiz deadlines (24 hours prior), and result publication.	Functionality	Medium	Student / Instructor	AC-CSQ-F-052: Notification is delivered within 1 minute of the triggering event. Test: TC-Notif-01	Depends on Notification Service
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4.7 Administration & Moderation

Requirements: Administrators shall manage user accounts and roles, review flagged/plagiarised content, and remove or suspend content and users that violate platform policy. IDs: CSQ-F-060..CSQ-F-062.

Req ID	Requirement	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
CSQ-F-060	The system shall allow an administrator to view, suspend, or reactivate a user account.	Functional	High	Administrator	AC-CSQ-F-060: A suspended account cannot log in; the action is recorded in the audit log. Test: TC-Admin-01	
CSQ-F-061	The system shall present administrators with a moderation queue of content flagged by users or the Plagiarism Detection Service.	Functional	High	Administrator	AC-CSQ-F-061: Flagged items appear in the queue with reason and source within 1 minute. Test: TC-Admin-02	Depends on Plagiarism Service
CSQ-F-062	The system shall allow an administrator to remove content or issue a warning to a user, with the action and reason recorded in the audit log.	Functional	High	Administrator	AC-CSQ-F-062: Removed content becomes inaccessible to students; an audit entry is created. Test: TC-Admin-03	Depends on CSQ-SR-005

5. Non-functional requirements (detailed)

NFRs below are measurable and tied to test plans. IDs CSQ-NF-###.

Req ID	Requirement	Category	Priority	Acceptance criteria / Measurement
CSQ-NF-001	95% of page loads (dashboard, material library, quiz player) shall complete within 3 seconds under normal load.	Performance	High	Load test shows p95 ≤ 3s. Test: TC-Perf-01
CSQ-NF-002	The platform shall provide 99.5% availability monthly, excluding scheduled maintenance windows.	Reliability	High	Uptime reports show ≥99.5% per month. Test: Ops reports
CSQ-NF-003	The system shall support at least 5,000 concurrent quiz attempts	Scalability	High	Load test at 5,000 concurrent users passes. Test: TC-Perf-02

Req ID	Requirement	Category	Priority	Acceptance criteria / Measurement
	without degradation beyond defined performance targets.			
CSQ-NF-004	All user activity (uploads, quiz attempts, admin actions) shall be logged with timestamps and retained for 2 years for audit purposes.	Audit / Data Retention	Medium	Automated archival and retrieval verified. Test: TC-OPS-01
CSQ-NF-005	The web interface shall conform to WCAG 2.1 AA accessibility guidelines.	Usability / Accessibility	Medium	Accessibility audit pass. Test: TC-UX-01

5.1 Security

5.1.1 Security Objectives

- Protect user credentials and personal data from unauthorized access or disclosure.
- Preserve academic integrity by preventing quiz-answer leakage and detecting plagiarised uploads.
- Ensure the integrity and availability of quiz results and leaderboard data.
- Maintain a tamper-evident audit trail of all administrative actions.

5.1.2 Security Requirements

Req ID	Requirement (shall...)	Type	Priority	Acceptance criteria / Test case ref
CSQ-SR-001	TLS 1.2+ mandatory for all network connections between client, server, and third-party services.	Security	High	TLS scan shows no weak ciphers. TC-Sec-01
CSQ-SR-002	User passwords shall be hashed using a salted, adaptive algorithm (e.g. bcrypt/argon2) and never stored in plaintext.	Security	High	Password store review confirms hashing. TC-Sec-02
CSQ-SR-003	Quiz questions and correct answers shall be encrypted at rest and only decrypted server-side at attempt time.	Security	High	Storage inspection confirms encryption. TC-Sec-03
CSQ-SR-004	The system shall enforce role-based access control (RBAC) so students cannot access instructor/admin functions.	Security	High	RBAC test matrix passes for all roles. TC-Sec-04
CSQ-SR-005	All administrative actions (user suspension, content removal) shall be logged with actor, timestamp, and reason.	Security	Medium	Audit log entries verified for sample actions. TC-Sec-05

6. Quality attributes & Acceptance tests

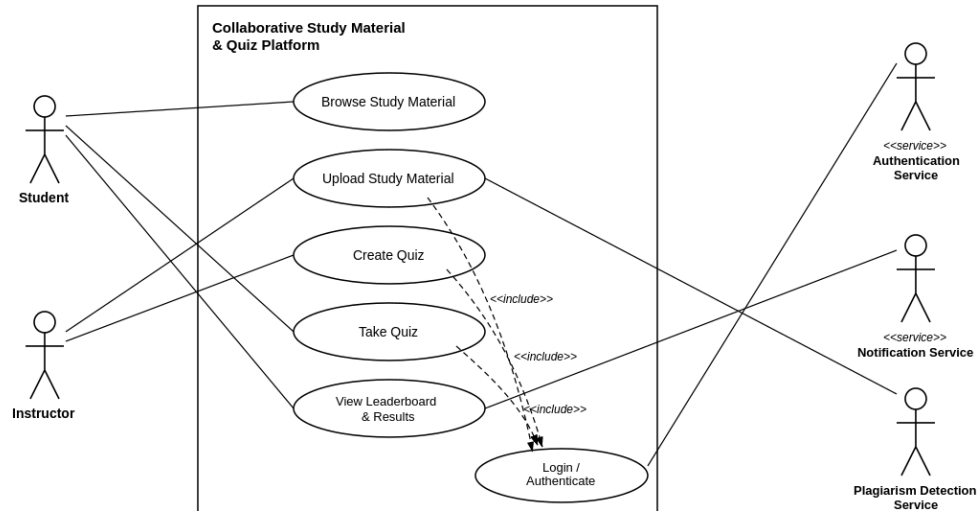
- Exit criteria for acceptance: All high-priority functional requirements implemented and verified, no critical NFR failures, and the RTM shows all mapped test cases passed.

- Acceptance test suites: Authentication, Material Management, Quiz Creation, Quiz Taking & Grading, Discussion & Rating, Notifications, Performance, Security, and Accessibility.

7. System models and diagrams

7.1 UML Use-Case diagram

The diagram below shows the primary actors (Student, Instructor) and external service actors (Authentication Service, Notification Service, Plagiarism Detection Service) interacting with the core use cases of the Collaborative Study Material & Quiz Platform. Taking a quiz, creating a quiz, and uploading study material all include the Login/Authenticate use case, since a verified session is required before these actions can be performed.



8. Requirements Traceability Matrix (RTM)

Req ID	Requirement short	Section ref / Design Spec	Module	Test case(s)	Status (N/P/A)	Comments
CSQ-F-001	User registration	4.1 / DS-Auth-01	AuthModule	TC-Auth-01	N	
CSQ-F-010	Upload material	4.2 / DS-Material-01	MaterialModule	TC-Mat-01, TC-Mat-02	N	
CSQ-F-030	Take quiz & auto-grade	4.4 / DS-Quiz-01	QuizModule	TC-Quiz-01	N	
CSQ-NF-001	Page load performance	5 / DS-Perf-01	WebUI / API Gateway	TC-Perf-01	N	
CSQ-SR-002	Password hashing	5.1.2 / DS-Sec-01	AuthModule	TC-Sec-02	N	